



STANDARD TIME — FULL MIND MAP

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1. Standard Meridian & Basic Facts

- India ka Standard Meridian **82°30' East** hai — isi meridian ke local time ko poore desh ka Indian Standard Time (IST) maana jaata hai. [Q1, Q10]
- Mainland India ka latitudinal aur longitudinal extent dono lagbhag **30°** hai. Gujarat se Arunachal Pradesh tak time-lag ~2 ghante ka hai — ढाई ghante NAHI (ye ek common wrong statement hai jo exam mein diya jaata hai). [Q1]

2. Standard Meridian Pass-Through States & Cities

- 82°30' E meridian **5 states** se guzarta hai: **Uttar Pradesh, Madhya Pradesh, Chhattisgarh, Odisha aur Andhra Pradesh** — Maharashtra se NAHI guzarta. [Q8]

EXAM TRAP — Mirzapur/Naini vs Allahabad — dono correct maane jaate hain

- 82°30' E meridian officially Uttar Pradesh ke **Mirzapur** district se guzarta hai — Prayagraj (Allahabad) ke paas. Isliye purane exams mein 'Prayagraj/Allahabad (Naini)' bhi correct answer maana jaata raha hai. In dono naamon ko interchangeably associate karo. [Q9, Q11]
- Rewa (M.P.) ka longitude (~81°19' E) diye gaye options mein IST meridian (82°30' E) ke sabse kareeb hai — Sagar, Ujjain, Hoshangabad se zyada close. [Q5]

3. IST vs GMT Time Difference

- IST, GMT (Greenwich Mean Time) se **+5½ ghante aage** hai. [Q12]

4. Time-Calculation Problems (Longitude ↔ Time)

- Calculation ka rule: **1° longitude = 4 minute time difference** (15° = 1 ghanta). Jo jagah IST meridian (82°30' E) ke east mein hai uska local time IST se **aage** hoga; jo west mein hai uska **peeche** hoga. [Q2, Q6, Q7, Q13]
- Example: Shillong (92°E) IST meridian se 9.5° east hai, isliye uska local time IST se 38 min aage hai (10:00 AM IST → 10:38 AM Shillong). Isi tarah Tirap (Arunachal) mein 5:00 AM sunrise ho to Kandla (Gujarat, ~30° west) mein sunrise ~7:00 AM hoga (2.5 ghante ka farak). [Q6, Q7, Q13]
- Reverse-calculation type: agar IST meridian par midnight (12 AM) ho aur kisi jagah 6 AM ho rahaa ho (6 ghante aage = 90° east), to wo jagah **82°30' E + 90° = 172°30' E** par hogi. [Q2]

- Gujarat ke westernmost village aur Arunachal Pradesh ke Walong (easternmost village) ke beech ka time-difference lagbhag **2 ghante** hai (longitude difference $\sim 29^{\circ}26'$). [Q3]

5. City Longitude Comparisons

- Jabalpur ka longitude ($\sim 79^{\circ}58'$ E) Indore ($75^{\circ}51'$ E) aur Bhopal ($77^{\circ}24'$ E) dono se zyada east hai — isliye Jabalpur 'inke beech mein' nahi aata. Aurangabad ka latitude ($\sim 19^{\circ}53'$ N) Vadodara aur Pune ke latitude ke beech mein aata hai. Bengaluru ($\sim 12^{\circ}58'$ N) Chennai ($\sim 13^{\circ}04'$ N) se zyada south hai. [Q4]
- Delhi ke sabse kareeb longitude wala city **Bengaluru** hai — Hyderabad, Nagpur, Pune se zyada close. [Q14]

6. Tropic of Cancer $\times$ IST Meridian Intersection

- Tropic of Cancer (23.5°N) aur IST meridian (82.5°E) ek dusre ko **Chhattisgarh ke Koriya district** mein cross karte hain, jo **Baghelkhand Plateau** region mein aata hai. [Q15]